Pankaj Rohilla, Ph.D.

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EDUCATION

Ph.D., Chemical Engineering, Texas Tech University *Aug 2017- Aug 2022*
M.Tech, Chemical Engineering, Indian Institute of Technology Kharagpur *Aug 2014- May 2016*
B.Tech, Chemical Engineering, Kurukshetra University *Aug 2009- May 2013*

EMPLOYMENT

Eckert Postdoc Fellow, Chemical and Biomolecular Engineering, **Georgia Tech** *Jun 2022-Present*
Adviser: Dr. Saad Bhamla and Dr. Mark Prausnitz
– *Development of low-cost hardware (electroporator) for intracellular delivery of mRNA and studying* in vivo *protein expression, cell viability immune response, and for vaccines and therapeutics.*
– *Evaluated the tolerability of the low-cost electroporator in human subjects.*
– *Interfacial locomotion of water-walking insects*
– *Viscoelastic jetting in tiny biological systems*
– *Fluid dynamics of flamingo feeding*
– *Fluid dynamics of recreational water sports*

Graduate Research Assistant, Chemical Engineering, **Texas Tech University** *2017- 2022*
Adviser: Dr. Jeremy Marston
– *Optimized the performance of needle-free jet injectors by understanding the hydrodynamics of jet injections*
– *Understanding the spreading dynamics of eye drops on front-of-the-eye.*
– *Studying the intradermal drug delivery via tattooing.*
– *Short-time dynamics of fluid-driven cracking.*

Project Officer, Chemical Engineering, **IIT Madras** *Dec 2016- Jun 2017*
Advisors: Dr. M.G. Basavaraj, Dr. S. Thampi, and Dr. M. Manivannan
– *Detection of critical micelle concentration via spreading oil drops on surfactant solutions*

Junior Research Fellow, Chemical Engineering, **IIT Bombay** *Jun 2016- Dec 2016*
Adviser: Dr. Jyoti Seth
– *Stochastic modeling of particle aggregation using Stokesian Dynamics*

Graduate Assistant, Chemical Engineering, **IIT Kharagpur** *Mar 2015- Jun 2016*
Adviser: Dr. Somenath Ganguly
– *Charge transport in carbon electrodes for supercapacitors*

AWARDS AND HONORS

Top 20 Scientific Contributions, *Controlled Release Society Meeting*, Bologna, Italy *2024*
Robert M. Nerem Travel Award, Georgia Tech *2023*
Eckert Postdoctoral Fellowship Award, Georgia Tech (ChBE) *2022*
Horn Distinguished Professors Graduate Achievement Award, Texas Tech University *2022*
Best Poster Award, Annual Chemical Engineering Research Fair, Texas Tech University *2022*
APS March FGSA Travel Award, Texas Tech University *2022*

Poster Award Winner, AICHE Fall Meeting (FP & BE Division) *2021*
Study Abroad Competitive Scholarship (SACS), Texas Tech University *Fall 2021*
Graduate School Travel Award, Texas Tech University *Fall 2019, 2021*
Best Judge, 3 minute presentation, Society of Plastics Engineers, Texas Tech *Spring 2021*
Mark Demark Scholarship, Texas Tech University *Spring 2021*
NSF I-Corps (Regional), Texas Tech University *Fall 2020*
Graduate Student Research Support Award, Texas Tech University *Spring 2020*
MHRD Scholarship, Indian Institute of Technology, Kharagpur *2014-2016*

PUBLICATIONS

Google Scholar

* *indicates equal contribution.*

16. **Rohilla, P.*** O'Neil, J.*, Jimenez, V. O., Choi, D., and Bhamla, M. S., **Interfacial vortex recapture enhances thrust in tiny water skaters**, **PNAS**, *In Revisions*, bioRxiv (2024).

15. **Rohilla, P.***, Choi, D.*, Wallace, H., Yung, K., Deora, J., Lele, A. and Bhamla, M. S. **Fluid dynamics of manu jumping: creating large splashes in water recreational sports**, Invited, **RSC Interface Focus**, bioRxiv, (2024).
Media: **New Scientist** , **CBC Radio**

14. Lu, C*, **Rohilla, P.***, Felner, E. I., Byagathvalli, G., Azizoglu, E., Bhamla,, M. S. and Prausnitz, M R. **Tolerability of a piezoelectric microneedle electroporator in human subjects**, **Bioengineering and Translational Medicine** 9 (4), e10662 (2024).

13. O'Neil, Johnathan, Yung, K. L., Difini, G., **Rohilla, P.**, and Bhamla, M. S. **Limb loss and specialized leg dynamics in tiny water-walking insects**, **Integrative and Comparative Biology** 64 (3), 1034-1043 (2024)

12. Challita, Ellio J., **Rohilla, P.**, and Bhamla, M. Saad. **Fluid ejections in nature**, **Annual Review of Chemical and Biomolecular Engineering**. *15* (2024).

11. **Rohilla, P.**, and Marston, J. O. **Focused high-speed liquid jets induced via low-voltage sparks in capillary tubes**, **Experiments in Fluids** 64 (5), 90 (2023).

10. **Rohilla, P.**, Khusnatdinov, E., and Marston, J. O. **Effect of air pockets in drug delivery via jet injections**, **International Journal of Pharmaceutics** 602, 120547 (2021).

9. Lawal, I., **Rohilla, P.**, and Marston, J. O. **Visualization of drug delivery via tattooing: effect of needle reciprocating frequency and fluid properties**, **Journal of Visualization**, 1-9 (2022).

8. Shahriar, M.*, Rewanwar, A.*, **Rohilla, P.***, and Marston, J. O. **Understanding the effect of counterpressure buildup during syringe injections**, **International Journal of Pharmaceutics** 602, 120530 (2021).

7. **Rohilla, P.**, and Marston, J. O. **Feasibility of laser induced jets in needle-free jet injections**, **International Journal of Pharmaceutics** 589, 119714 (2020).
Media: **New Scientist**

6. **Rohilla, P.**, Lawal, I., Blanc, A.L., O'Brien, V., Weeks, C., Tran, W., Rane, Y.S., Khusnatdinov, E., and Marston, J.O. **Loading effects on the performance of needle-free jet injections in different skin models**, **Journal of Drug Delivery Science and Technology** 60, 102043 (2020).

5. Deodhar, S., **Rohilla, P.**, Manivannan, M., Thampi, S.P., and Basavaraj, M.G. **Robust method to determine critical micelle concentration via spreading oil drops on surfactant solutions**, Langmuir 36 (28): 8100-8110 (2020).

4. **Rohilla, P.**, Rane, Y.S., Lawal, I., Blanc, A.L., Davis, J., Thomas, J.B., Weeks, C., Tran, W., Fisher, P., Broderick, K.E., Simmons, J.A. and Marston J.O., **Characterization of jets for impulsively-started needle-free jet injectors: Influence of fluid properties**, Journal of Drug Delivery Science and Technology 53, 101167 (2019).

3. **Rohilla, P.** and Marston, J. O. **In-vitro studies of jet injections**, International Journal of Pharmaceutics 568, 118503 (2019).

2. **Pankaj**, Chavhan, M.P. and Ganguly, S., **Charge transport in activated carbon electrodes: the behaviour of three electrolytes vis-à-vis their specific conductance**, Ionics 23, 2037 (2017) .

1. Chavhan, M.P., **Pankaj** and Ganguly, S. **Charge transport in carbon electrodes made by electrospray of precursor sol and subsequent carbonization in situ**, Journal of Solid State Electrochemistry 22, 7 : 2149-2157 (2018).

WORK IN PROGRESS

∗ indicates equal contribution.

5. Ortega, V.M. Tien, **Rohilla, P.**, Seleb, B.R., Belair, J., and Bhamla, M. S. Flamingos use L-shaped beak and morphing feet to induce vortical traps for prey capture, **PNAS**, *Under Review*.

4. Challita, E., Harrison, J., **Rohilla, P.**, and Bhamla, M. S. Viscoelastic jets from ultrasmall nozzles in termites. (*In Preparation*).

3. **Rohilla, P.**, Lawal, I., Williams, N., and Marston, J. O. Early-time dynamics of fluid driven cracks, *In submission*

2. **Rohilla, P.**, Williams, N., and Marston, J. O. Fluid driven cracking in multilayered hydrogels with high-speed liquid jets, *In submission*

1. Lawal, Idera, **Rohilla, P.**, Rodriguez, E., Pham, P. and Marston, J. O. Droplet spreading on eye-like substrates, *In submission*

FUNDING

Total Funds Raised: > **$ 210,000**

1. Eckert Postdoctoral Fellowship Award: $110,000
Won the competitive fellowship to obtain funding for 2 years towards monthly salary and travel funds.

2. Georgia Research Alliance Grants - Phase I & II (PIs: *Saad Bhamla & Mark R. Prausnitz*): >$100,000
Drafted grant proposal and reports.

PRESENTATIONS

Invited

4. **Keynote Lecture - 4th International Conference on Future Technologies in Manufacturing, Automation, Design & Energy (NIT Trichy, India)**, *Fluid ejections in nature* (Dec 2024)

3. **University of Alabama (Tuscaloosa, AL)**, Chemical and Biological Engineering *Addressing global health challenges using high-speed liquid jets and ultra-low-cost tools* (Jan 2024)

2. **Georgia Institute of Technology (Atlanta, US)**, Quantitative Biosciences, *Principles of locomotion - Water walkers* (Nov 2023)

1. **Karolinska Institutet (Stockholm, Sweden),** April 2023. *Ultra-low-cost electroporator for intradermal delivery of nucleic acids.* (Apr 2023)

Contributed

20. **American Institute of Chemical Engineers - Annual Fall Meeting**, San Diego, CA, 2024. "*Epatch: An Ultra-Low-Cost Handheld Electroporator for Intradermal Delivery of mRNA.*" (Oral)

19. **American Institute of Chemical Engineers - Annual Fall Meeting**, San Diego, CA, 2024. "*Vortical Interactiona in nature.*" (Oral)

18. **American Physical Society - March Meeting**, Minneapolis, MN, 2024, "*Vortical interactions in nature.*" (Oral)

17. **American Physical Society - March Meeting**, Minneapolis, MN, 2024, "*Vortex interactions in Water-walking insects.*" (Poster)

16. **The Society for Integrative and Comparative Biology - Annual Meeting**, Seattle, WA, 2023, "*Small yet fast water-walkers: vortex interactions during water locomotion in Microvelia.*" (Oral)

15. **American Physical Society - Division of Fluid Dynamics Meeting**, Washington DC, 2023. "*Studying vortex interactions in water walking insects using physical and computational fluid dynamics.*" (Oral)

14. **American Institute of Chemical Engineers - Annual Fall Meeting**, Orlando, FL, 2023. "*Electroporation-Mediated Delivery of mRNA in the Skin Using a Low-Cost Handheld Electroporator.*" (Oral)

13. **American Physical Society - March Meeting**, Las Vegas, NV, 2023. "*Impact of vortex recapture in water-walking Microvelia using a physical model and computational fluid dynamics.*" (Oral)

12. **The Society for Integrative and Comparative Biology - Annual Meeting**, Austin, TX, 2023, "*Physical and computational models of vortex recapture during Microvelia's walking on water.*" (Oral)

11. **American Physical Society - March Meeting**, Chicago, IL, 2022, "*Spark-induced drops and jets.*" (Oral)

10. **CHEGSA Symposium - Tech University**, Lubbock, TX, 2022, "*Optimizing needle-free jet injections for intradermal drug delivery.*" (Poster). **First Prize.**

9. **Graduate School Symposium - Tech University**, Lubbock, TX, 2022, "*Optimizing needle-free jet injections for intradermal drug delivery.*" (Poster)

8. **American Physical Society - Division of Fluid Dynamics Meeting**, Phoenix, AR, 2022. "*Early-time dynamics of fluid-driven cracks.*" (Oral)

7. **American Institute of Chemical Engineers - Annual Fall Meeting**, Boston, MA, 2021. "*Optimizing needle-free jet injections for intradermal drug delivery.*" (Poster). **Best Poster Award.**

6. **American Institute of Chemical Engineers - Annual Fall Meeting**, Boston, MA, 2021. "*Early-time dynamics of fluid-driven cracks.*" (Oral)

5. **Graduate School Symposium - Texas Tech University**, Lubbock, TX, 2021, "*Laser-induced jets for drug delivery.*" (Poster)

4. **American Physical Society - Division of Fluid Dynamics Meeting**, Seattle, WA, 2019. "*Effect of applied load and jet dispersion on efficiency of needle-free injections.*" (Oral)

3. **American Physical Society - Division of Fluid Dynamics Meeting**, Atlanta, GA, 2018. "*In-vitro studies of jet injection dynamics.*" (Oral)

2. **International Conference on Material Science and Engineering**, Kottayam, India, 2016. "*Impedance Spectroscopy Studies for Supercapacitors based on different electrolytes.*" (Poster).

1. **Annual Session of Indian Institute of Chemical Engineers**, IIT Guwahati, India, 2015. "*Modeling of Electric Double layer Capacitors.*" (Poster).

TEACHING EXPERIENCE

3. **Advanced Chemical Engineering Techniques, CHE 5310** (Teaching Assistant) *Fall 2018*
2. **Engineering Materials Science, CHE 3330** - (Teaching Assistant) *Spring 2018*
1. **Chemical Engineering Thermodynamics II, CHE 3322** - (Teaching Assistant) *Fall 2017*

MENTORSHIP

– Individually supervised **26** high school$^{\alpha}$, undergraduate$^{\beta}$ and graduate students$^{\gamma}$ in summer and semester-long research projects.
– †Co-authors in peer-reviewed publications.
– ‡Won the President's Undergraduate Research Award (Georgia Tech) and the Undergraduate Research Award (Texas Tech) in my mentorship.

26. **Sion Park**$^{\beta,\ddagger}$	Georgia Tech	*Fall 2024- Present*
25. **Jace Holmes**$^{\beta}$	Georgia Tech	*Fall 2024- Present*
24. **Sarah Bender**$^{\beta}$	Georgia Tech	*Fall 2024- Present*
23. **Avaneesh Choragudi**$^{\alpha}$	Lambert High School	*Fall 2024- Present*
22. **Vedant Mehta**$^{\alpha}$	Lambert High School	*Fall 2024- Present*
21. **Atharva Lele**$^{\beta,\ddagger}$	Georgia Tech	*Fall 2023- Present*
20. **Annika Joshi**$^{\alpha}$	Johns Creek High School	*Fall 2023- Summer 2024*
19. **Johnathan O'Neil**†	Georgia Tech	*2022-2024*
18. **Holden Walker**$^{\beta,\dagger,\ddagger}$	Georgia Tech	*Fall 2022-Spring 2023*
17. **Nihanth Pinakka**$^{\beta}$	Georgia Tech	*2022-2023*
16. **Breanna Carruth**$^{\beta}$	Texas Tech University	*2021-2022*
15. **Eliana Rodriguez**$^{\beta}$	Texas Tech University	*Spring 2022*
14. **Elina Khusnatdinov**$^{\beta}$	Texas Tech University	*2021- 2022*
13. **Emil Khusnatdinov**$^{\beta,\dagger}$	Texas Tech University	*2020- 2022*
12. **Noah Williams**$^{\beta}$	Texas Tech University	*2020- 2022*
11. **Md. Shahriar**†	Texas Tech University	*2019- 2020*
10. **Ankit Rewanwar**†	Texas Tech University	*2019- 2020*
9. **Cormak Weeks**$^{\beta,\dagger,\ddagger}$	Texas Tech University	*2019- 2022*

8. **Whitney Tran**[β,†]	Texas Tech University	*2019- 2022*
7. **Veronica O'Brien**[α,†]	Margaret Talkington School	*Summer 2019*
6. **Pedro Mallet**	Fed. Flum. University, Brazil	*Summer 2019*
5. **Andrew Le-Blanc**[β,†]	Texas Tech University	*Spring 2019*
4. **Justin Davis**[β,†]	Texas Tech University	*2019*
3. **Idera Lawal**[†]	Texas Tech University	*2018-2019*
2. **James B. Thomas**[β,†]	Texas Tech University	*2018*
1. **Haley Slook**[β]	Texas Tech University	*Fall 2018*

SERVICE

SESSION CHAIR of the following:
- American Physical Society March Meeting (2023)
- American Physical Society - Division of Fluid Dynamics Meeting (2023)
- Society of Integrative and Comparative Biology Annual Meeting (2024)

LEADERSHIP in the following positions:
- Vice-President, Chemical Engg. Graduate Student Association, Texas Tech (2018-2019)
- Graduate Student Rep., Chemical Engg. Student Advisory Council, Texas Tech (2018-2019)

REVIEWER in the following peer-reviewed journals:
- Bioengineering and Translational Medicine
- Journal of Drug Delivery Science and Technology
- Journal of Heat and Mass Transfer
- Scientific Reports
- Royal Society Proceedings B
- HardwareX.

VOLUNTEER in the following outreach activities:
- Zoo Biomechanics Day, Atlanta (2023, 2024)
- Atlanta Science Festival (2024)
- Workshops in Peru (Rural and Local Girls' Primary Schools, Porto Maldonaldo and Lima) (2022)
- Annual Science Fair, Lubbock, TX (2019)

PROFESSIONAL MEMBERSHIPS

MEMBER of the following societies:
- American Institute of Chemical Engineers (AICHE)
- American Physical Society (APS)
- The Society of Integrative and Comparative Biology (SICB)
- Society of Plastic Engineers (SPE)

REFERENCES

Saad Bhamla, Assoc. Professor, Georgia Institute of Technology saadb@chbe.gatech.edu
Mark R. Prausnitz, Professor, Georgia Institute of Technology prausnitz@gatech.edu
Jeremy O. Marston, Assoc. Professor, Texas Tech University jeremy.marston@ttu.edu